Handheld: pairing and warehouse work

This is an informational translation. The matching Russian revision is authoritative. Step-by-step warehouse instruction for Markiro Autonom: pairing a handheld terminal with the host by a code from the office, operator sign-in, the label printer, building a pallet from boxes, printing the pallet label, and working without a link to the host.

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1. Purpose

This instruction covers the warehouse «Паллеты» (Pallets) mode on the handheld terminal (ТСД) in Markiro Autonom. The operator scans finished boxes' SSCC labels; the handheld closes the pallet and prints its label on the assigned printer. Production-shift work on the handheld is outside this instruction's scope. Line-station work is covered in MKR-INS-01–03.

The handheld works with the host, the site's main computer, over the local network. No internet connection is needed. While the host is unavailable, the handheld keeps working and sends what it has collected once the link returns.

Note: The Markiro Autonom interface is in Russian only, so the screenshots are Russian and screen labels are quoted in Russian with an English translation in brackets. The screenshots use demonstration data; addresses, codes, the host fingerprint and SSCC numbers on your site will differ.

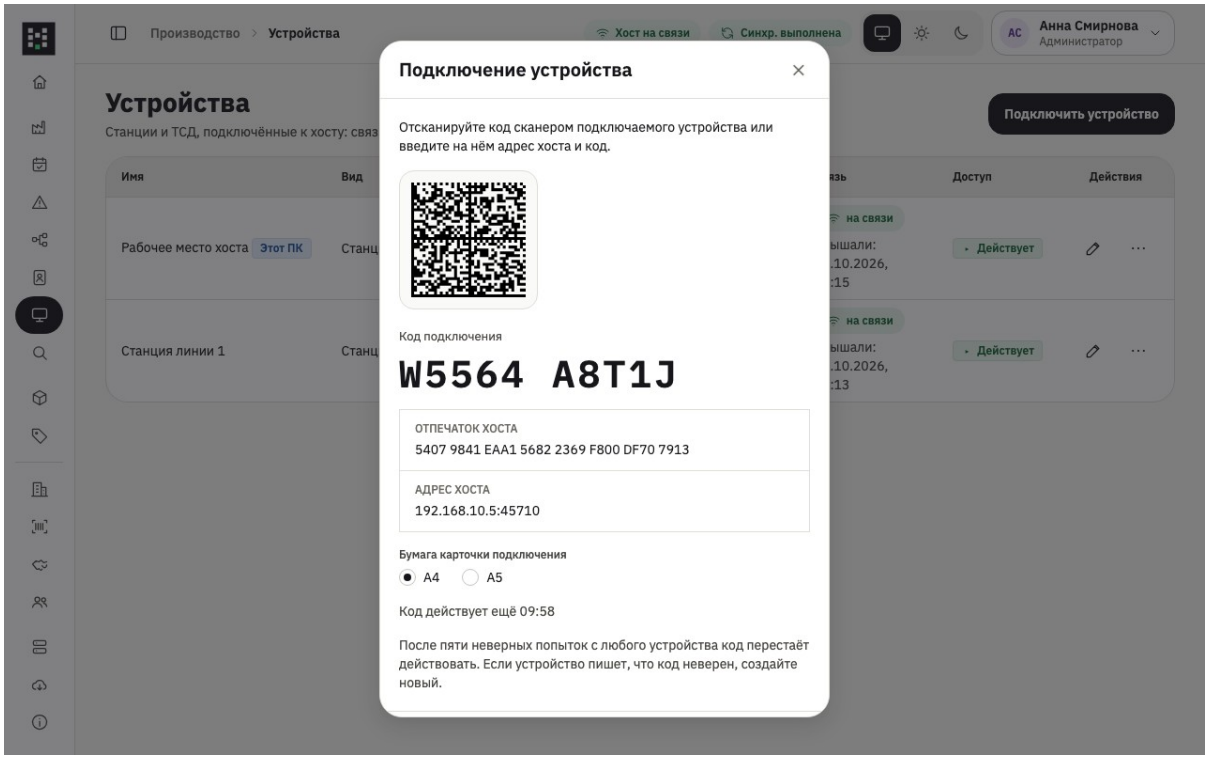
2. Preparation

- The host is installed and running, and the handheld is on the same Wi-Fi network as the host computer.
- The host computer has a permanent network address: the handheld remembers the address when it is paired. Reserving an address is described in the host installation instruction.
- On the host computer the Wi-Fi network is marked as private, and client isolation is off on the access point; otherwise the handheld cannot see the host although both are on the network.
- The clocks of the handheld and the host computer agree within two minutes.
- You have access to the Markiro Autonom office as an administrator: the pairing code is created in «Устройства» (Devices).
- The label printer is reachable over Wi-Fi or Bluetooth, and you know its address, language (ZPL or TSPL) and resolution.

3. Pairing the handheld with the host

Step 1. 3.1. Create a code in the office

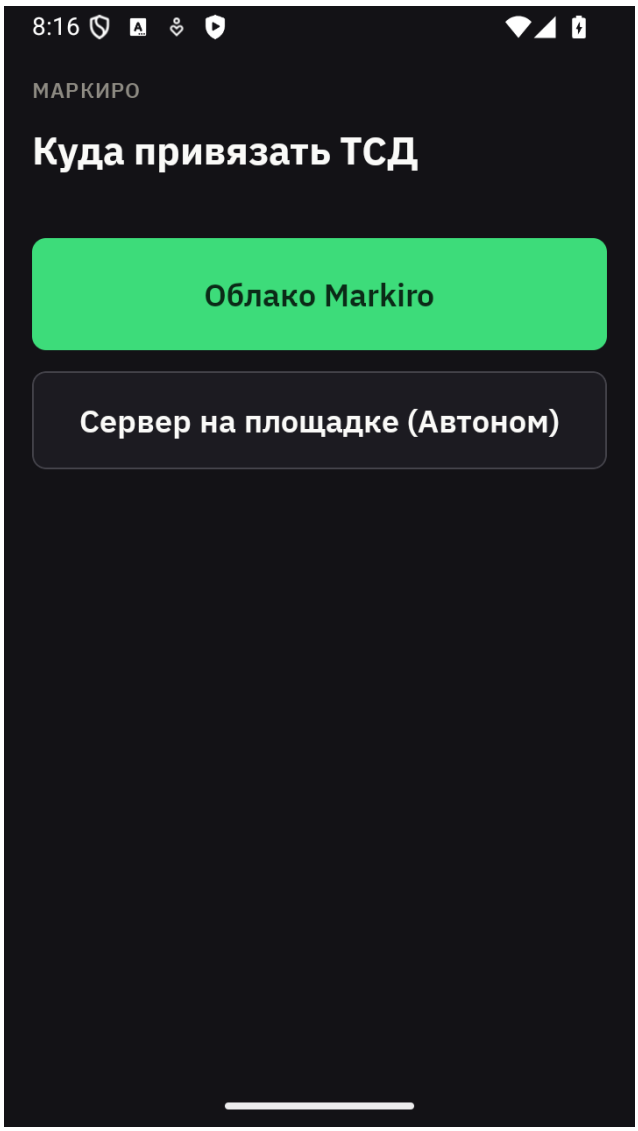
In the office open «Устройства» (Devices) and press «Подключить устройство» (Connect a device). The office shows the pairing barcode, the 10-character «Код подключения» (Pairing code), and the host's fingerprint and address. The code is single-use; the time left is shown under it, and «Новый код» (New code) issues another one.



Office: the pairing code for a new device

Step 2. 3.2. Choose the on-site server

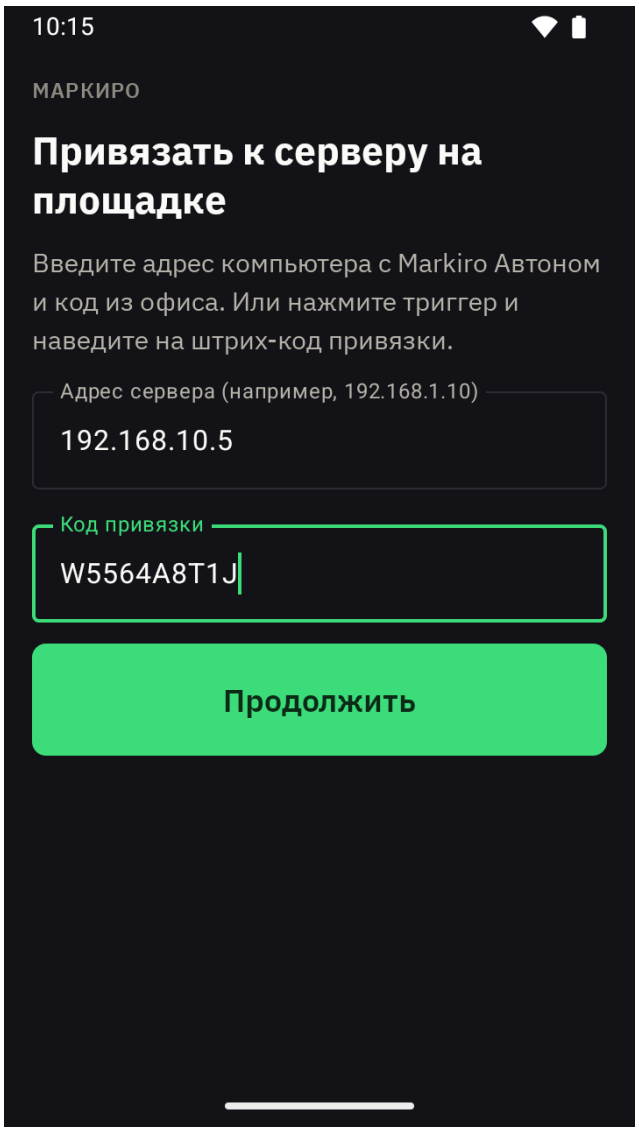
Start the app on the handheld. On «Куда привязать ТСД» (Where to pair the handheld) press «Сервер на площадке (Автоном)» (On-site server (Autonom)).



Handheld: choosing where to pair the device

Step 3. 3.3. Scan the barcode or type the address and code

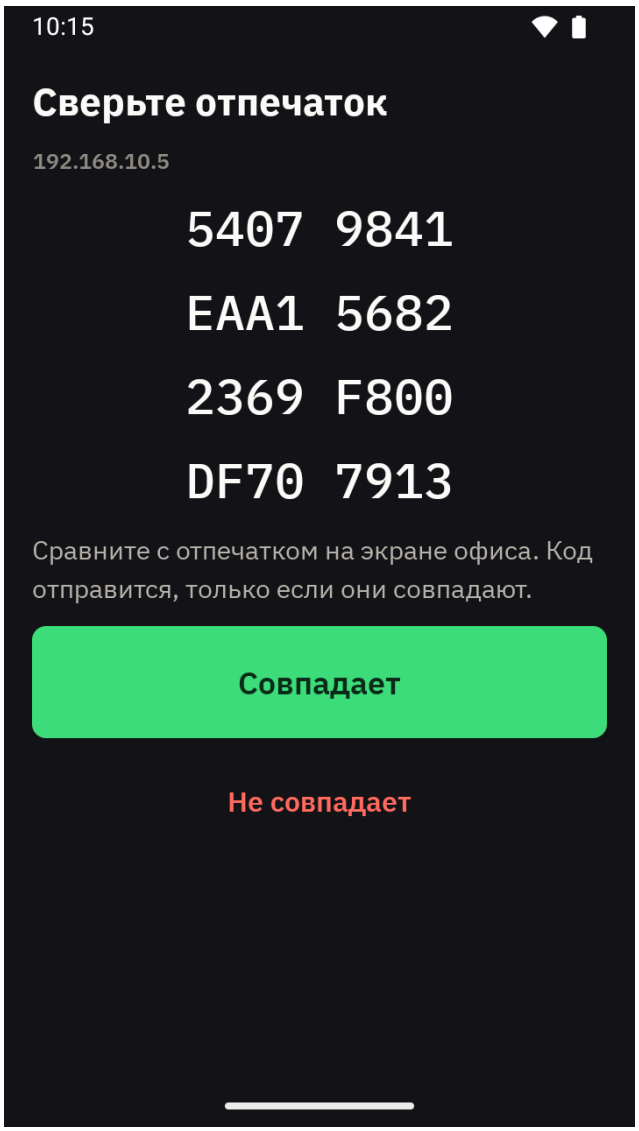
The simplest way is to press the trigger and aim the handheld at the pairing barcode on the office screen. Otherwise type «Адрес сервера» (Server address), the host address from the office without the port, for example 192.168.1.10, and «Код привязки» (Pairing code), then press «Продолжить» (Continue).



Handheld: the host address and pairing code typed by hand

Step 4. 3.4. Compare the fingerprint

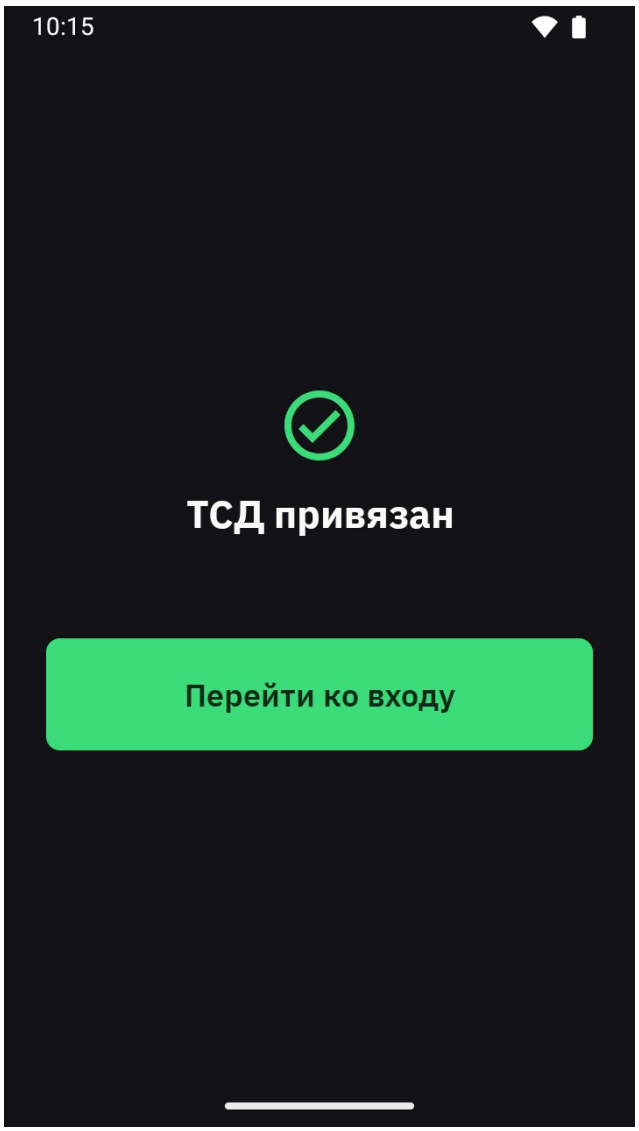
After manual entry the handheld shows «Сверьте отпечаток» (Compare the fingerprint) with the fingerprint of the host at that address. Compare it with the fingerprint on the office screen: the code is sent to the host only if they match. Press «Совпадает» (Matches); if the fingerprint differs, press «Не совпадает» (Does not match) and check the address. When the barcode is scanned, the handheld checks the fingerprint itself.



Handheld: comparing the host fingerprint

Step 5. 3.5. Wait for pairing to finish

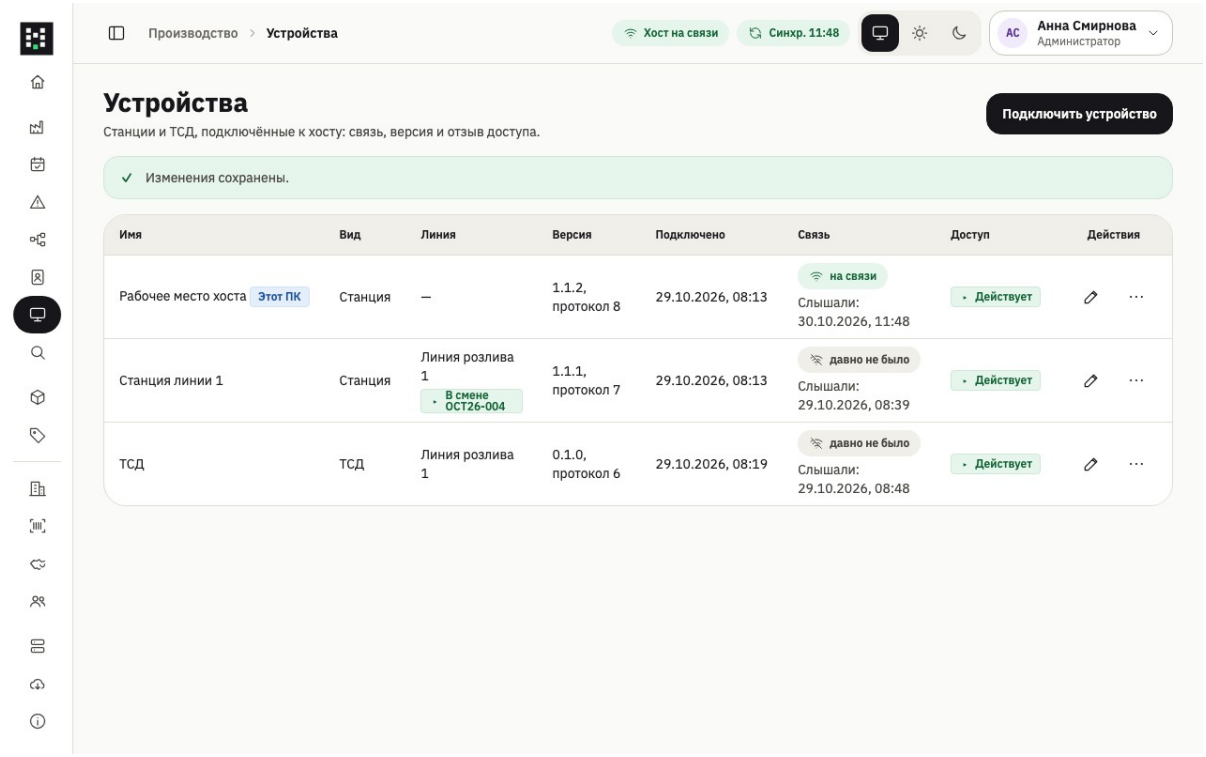
The handheld loads the reference data from the host and shows «ТСД привязан» (Handheld paired). Press «Перейти ко входу» (Go to sign-in).



The handheld is paired with the host

Step 6. 3.6. Check the device in the office

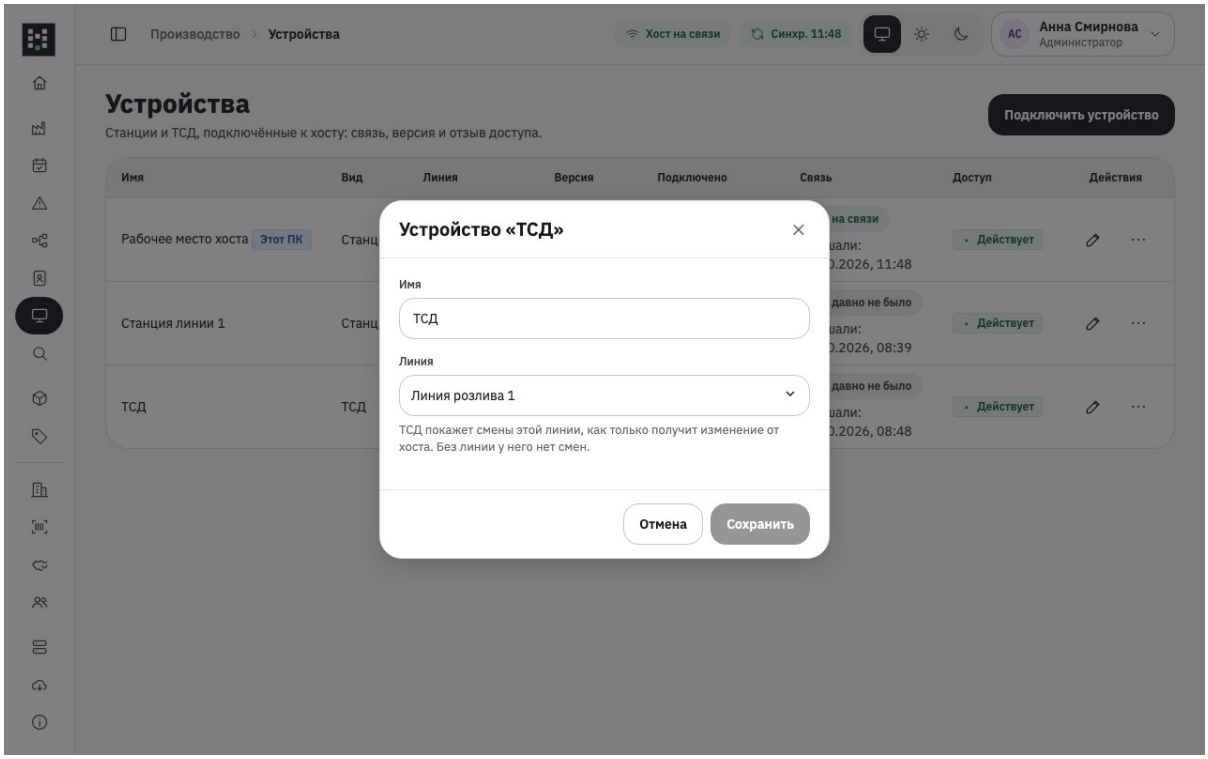
The office shows «Устройство «ТСД» подключено.» (Device “ТСД” connected.), and the «Устройства» (Devices) list gets a row of kind «ТСД» (Handheld). The device name can be changed with the pencil button in its row.



Office: the handheld in the device list

Step 7. 3.7. Assign a line if needed

If the handheld will work in a production shift, the administrator presses the pencil in its device row, chooses «Линия» (Line), and saves the change. The handheld receives the line when connected to the host; the «Смена» (Shift) tile lists that line's planned and active shifts. The warehouse «Паллеты» (Pallets) mode needs no line. If the «Линия» field is absent, update the office PC application to a version that supports handheld line assignment.



Office: assigning the handheld's line

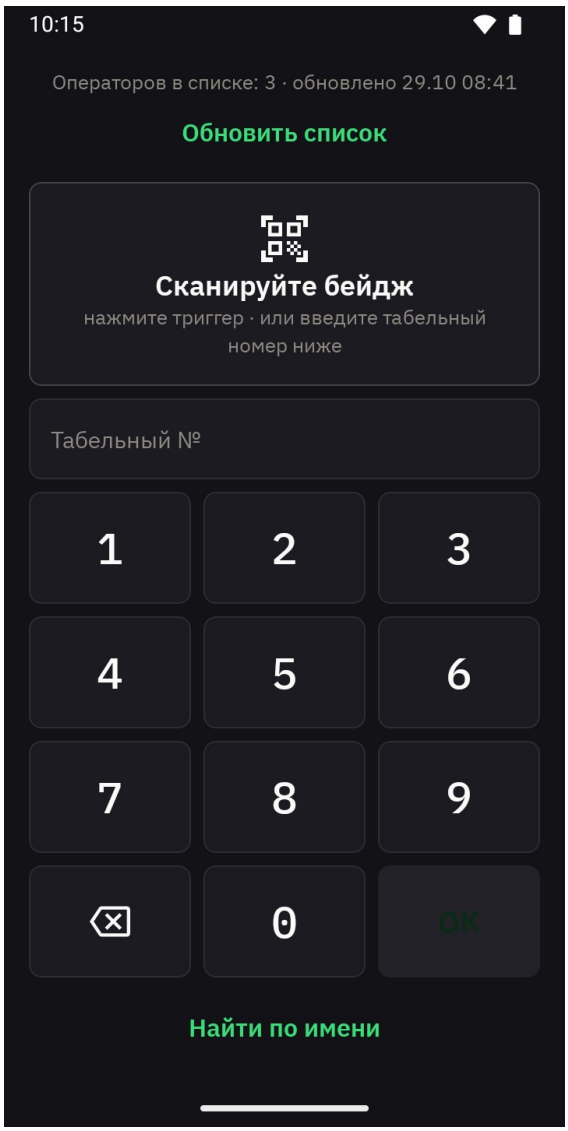
Message on the handheld	What to do
«Срок кода истёк. Выпустите новый код в офисе.» (The code has expired. Issue a new code in the office.)	In the office press «Новый код» (New code) and pair again.
«Код не подошёл. Выпустите новый код в офисе.» (The code did not match. Issue a new code in the office.)	Check the code against the office screen or issue a new one.
«Сервер не отвечает…» (The server does not respond…)	Check Wi-Fi on the handheld, the host address, the network profile and client isolation on the access point.
«В лицензии нет свободных мест для ТСД.» (The licence has no free handheld seats.)	The licence sets the number of handhelds; contact the administrator.
«Работа с ТСД не включена в офисе.» (Handheld work is not enabled in the office.)	Contact the administrator.

If pairing fails

4. Operator sign-in

Step 1. Sign in with a badge or personnel number

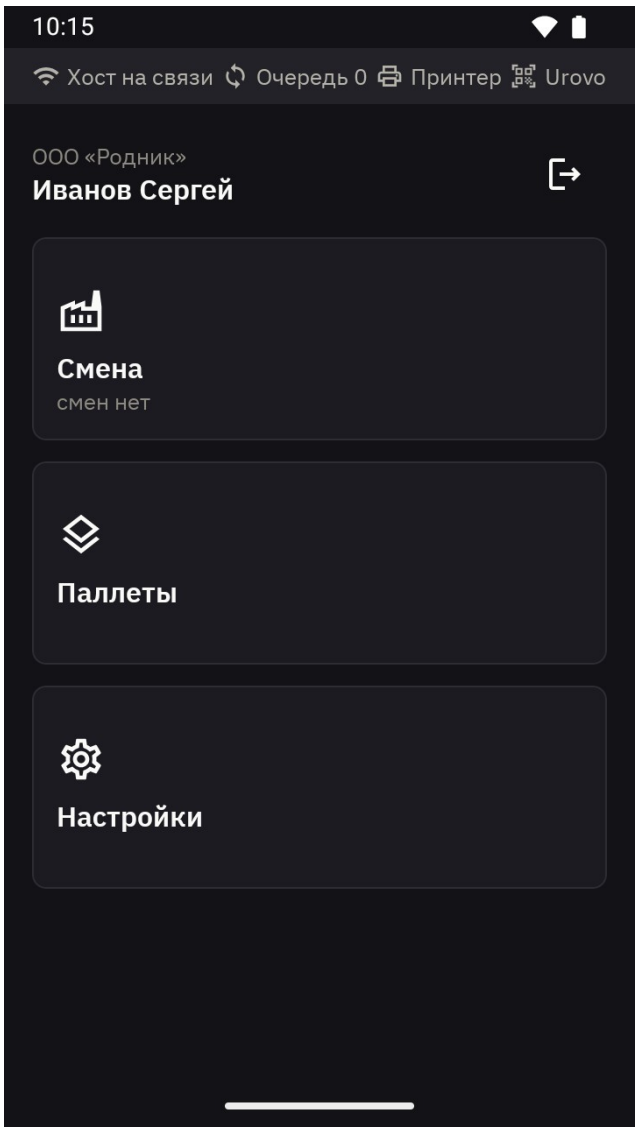
On the sign-in screen scan your badge, or type «Табельный №» (Personnel No.) on the on-screen keypad and then your PIN. If you do not remember the number, press «Найти по имени» (Find by name). The operator list is stored on the handheld, so sign-in also works without a link to the host; the line at the top shows how many operators are in the list and when it was updated, and «Обновить список» (Refresh list) loads it again.



Handheld: operator sign-in

Step 2. The main screen

After sign-in the handheld shows its main screen. The status line shows the link to the host («Хост на связи», Host online), the «Очередь» (Queue) of unsent records, the printer and the scanner. The tiles are «Смена» (Shift), «Паллеты» (Pallets) and «Настройки» (Settings). The exit button to the right of the name ends the operator's session.

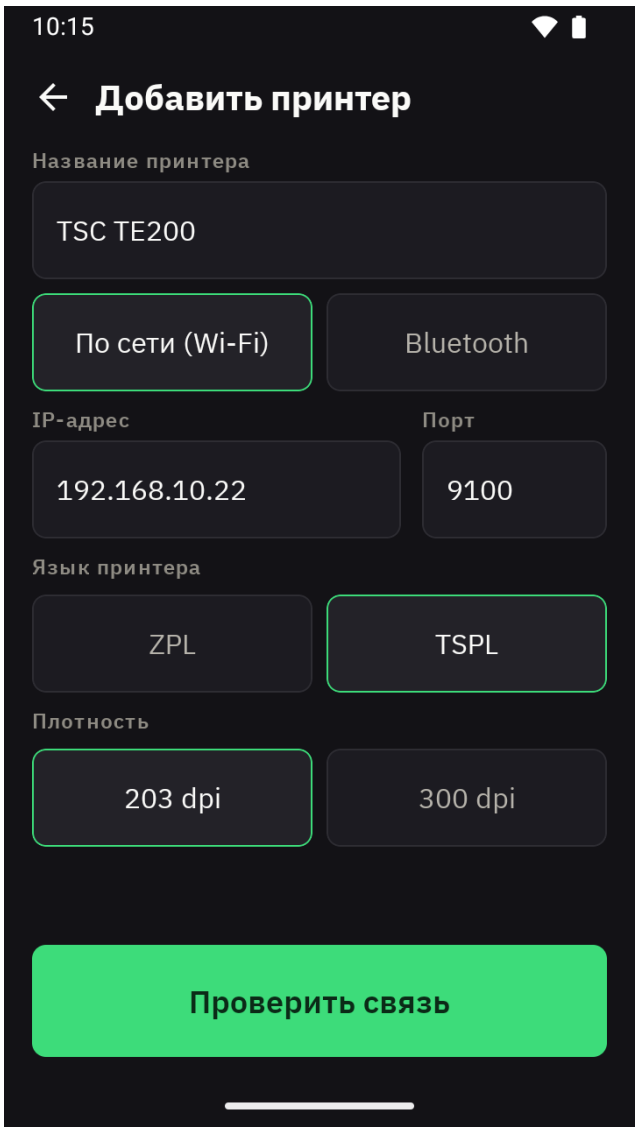


Handheld: the main screen without an assigned line

5. The label printer

Step 1. Add the printer

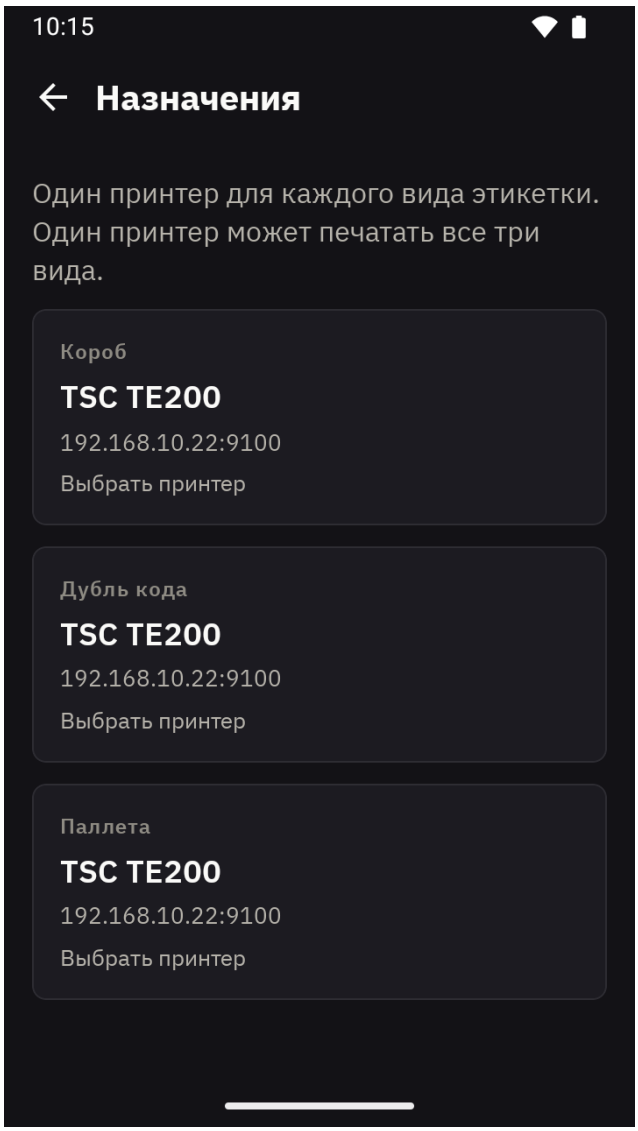
Open «Настройки» (Settings) → «Принтер» (Printer) and press «Добавить по адресу» (Add by address). Set «Название принтера» (Printer name), choose «По сети (Wi-Fi)» (Network (Wi-Fi)) or «Bluetooth», and for a network printer «IP-адрес» (IP address) and «Порт» (Port), usually 9100. Choose «Язык принтера» (Printer language): ZPL for Zebra printers, TSPL for TSC printers, and «Плотность» (Density), 203 or 300 dpi, as on the printer. Press «Проверить связь» (Check connection): the printer is saved on the handheld.



Handheld: a new network printer

Step 2. Assign the printer

On «Принтеры» (Printers) press «Назначения» (Assignments), and for each label kind, «Короб» (Box), «Дубль кода» (Code duplicate) and «Паллета» (Pallet), press «Выбрать принтер» (Choose printer). The warehouse needs the «Паллета» (Pallet) assignment. While a kind has no printer, the «Настройки» (Settings) tile on the main screen says what is not assigned.

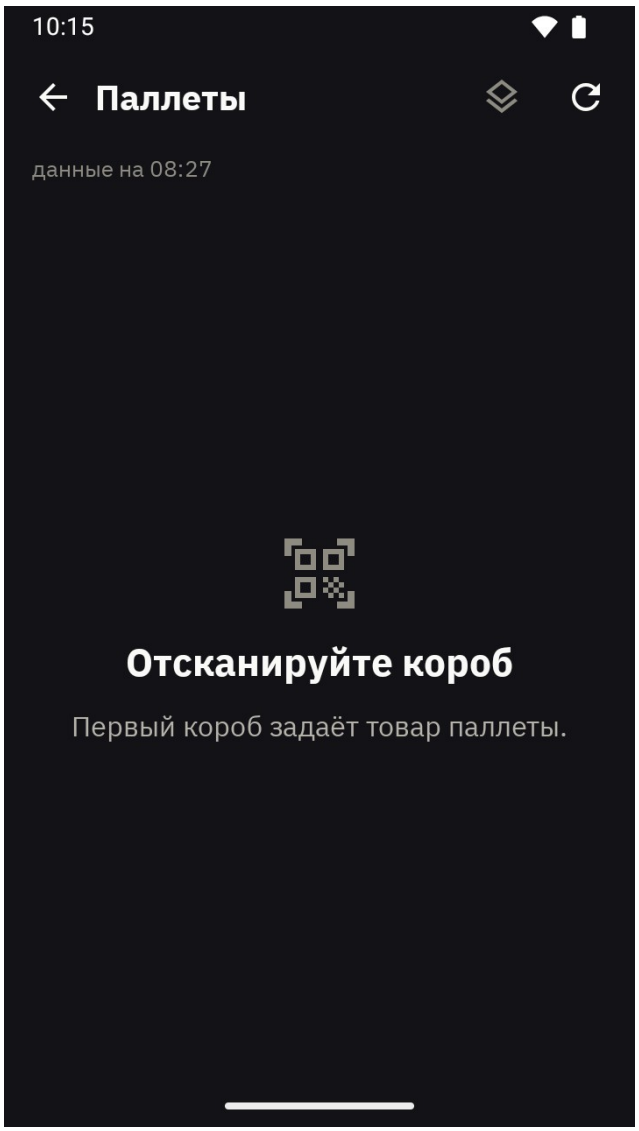


Handheld: printer assignment by label kind

6. Building a pallet

Step 1. Open pallet building

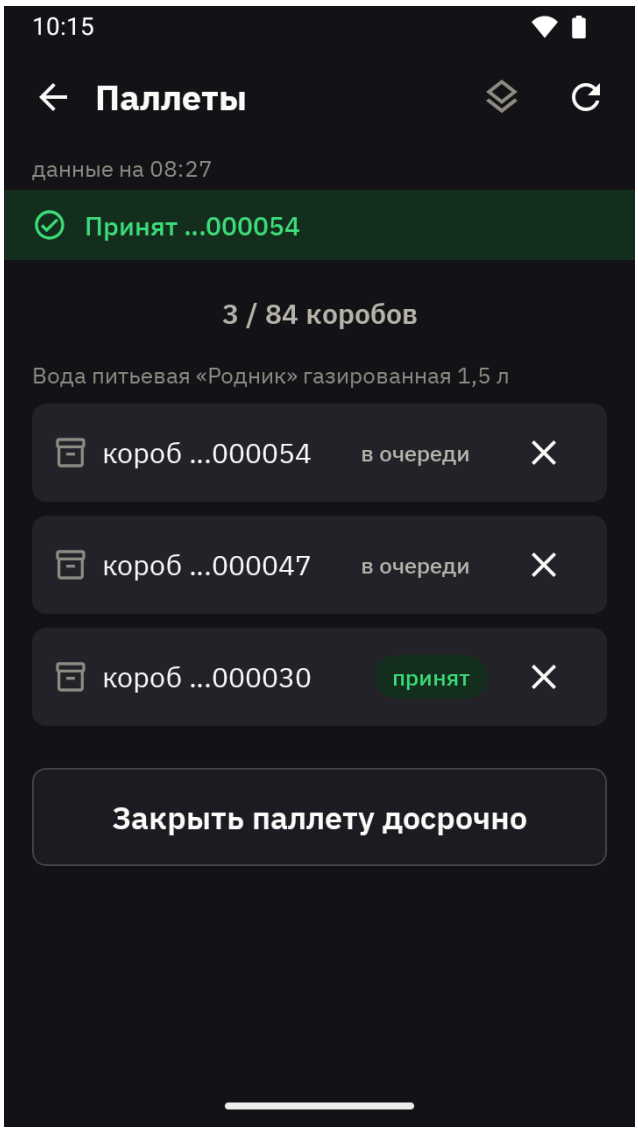
On the main screen press «Паллеты» (Pallets). The handheld waits for the first box: «Отсканируйте короб» (Scan a box). The first box sets the pallet's product. The line «данные на …» (data as of …) shows when the handheld last loaded the box registry from the host; «Обновить реестр» (Refresh registry) at the top of the screen loads it again.



Handheld: a new pallet waits for the first box

Step 2. Scan the boxes' SSCC labels

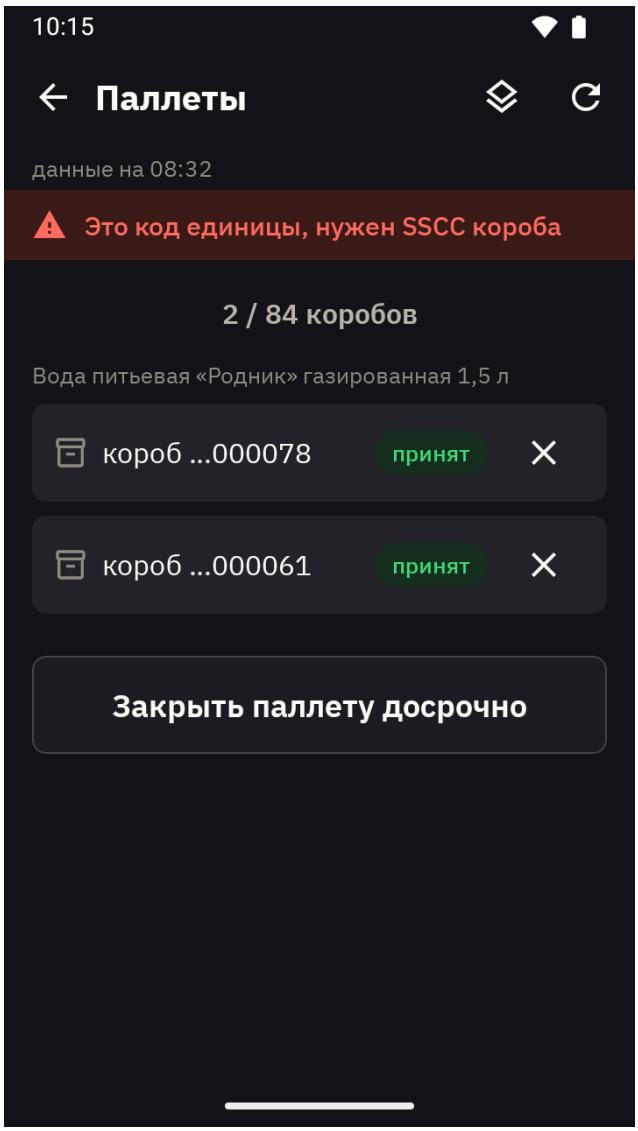
Scan the box labels one at a time. An accepted box appears in the list, the top line says «Принят» (Accepted) with the last digits of the SSCC, and above the list the handheld shows how many boxes are on the pallet out of the pallet capacity in the product card. Each box is first «в очереди» (queued) and then «принят» (accepted) by the host. A box put on by mistake is removed with the cross button in its row.



Handheld: three boxes on the pallet

Step 3. If a box is not accepted

When a scan does not fit, the handheld gives the reason in a red line at the top and does not put the box on the pallet. The most common messages are in the table below.



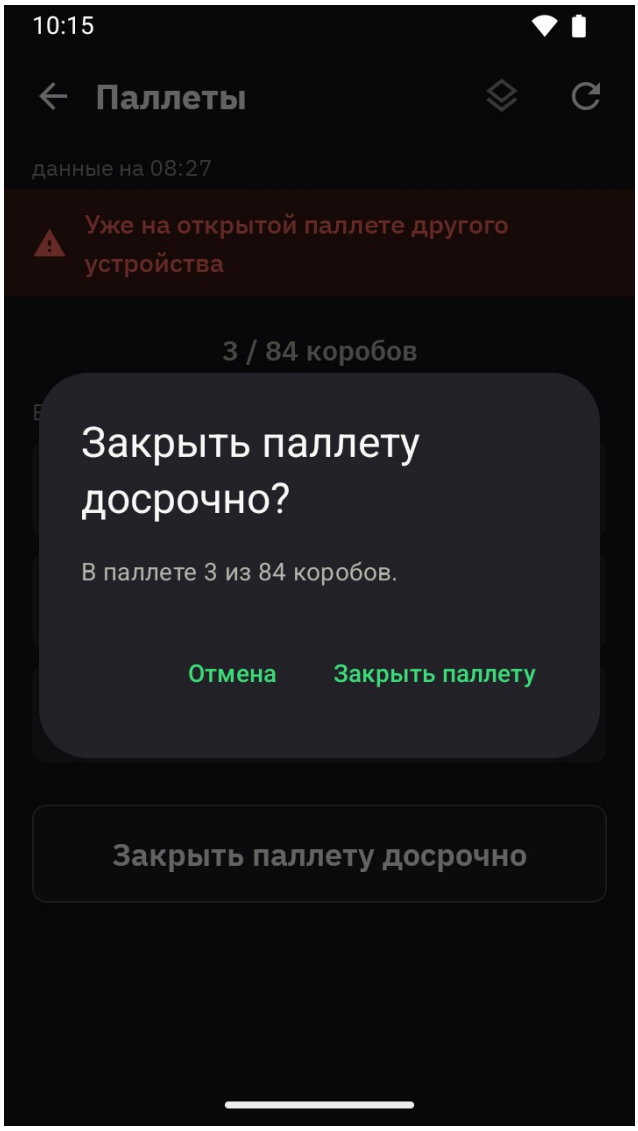
Handheld: a unit code scanned instead of a box SSCC

Message	What to do
«Уже на этой паллете ...» (Already on this pallet ...)	Nothing: the box is already in the list.
«Уже на паллете ...» (Already on pallet ...), «Уже на открытой паллете другого устройства» (Already on another device's open pallet)	The box is recorded on another pallet. Set it aside and tell your supervisor.
«Уже на другой паллете этого ТСД» (Already on another pallet of this handheld)	The box is on another pallet of this handheld. Set it aside and tell your supervisor.
«Другой товар: ...» (Another product: ...)	A pallet takes boxes of one product. Set the box aside.
«Короб неизвестен. Обновите реестр» (Unknown box. Refresh the registry)	Press «Обновить реестр» (Refresh registry) and scan the box again.
«Это код единицы, нужен SSCC короба» (This is a unit code; a box SSCC is needed)	Scan the box label, not the code on a unit.
«Это паллета, не короб» (This is a pallet, not a box)	A pallet label was scanned. Scan the box label.

Messages when a box is scanned

Step 4. Close the pallet

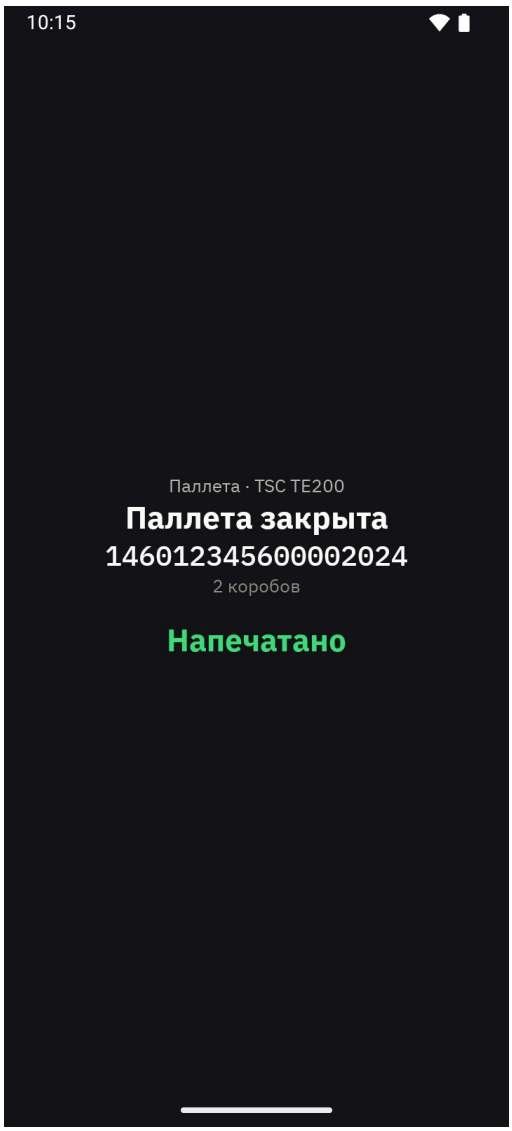
When the number of boxes reaches the pallet capacity, the handheld closes the pallet itself. To close earlier, press «Закрыть паллету досрочно» (Close pallet early) and confirm with «Закрыть паллету» (Close pallet).



Handheld: closing a pallet early

Step 5. Apply the pallet label

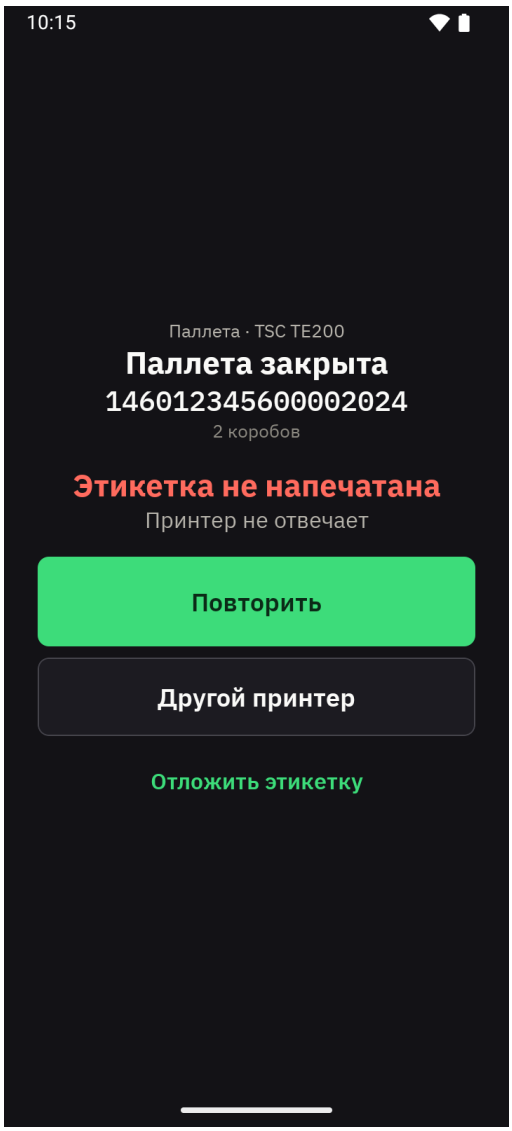
After closing, the handheld prints the pallet label on the printer assigned to «Паллета» (Pallet): the «Паллета закрыта» (Pallet closed) screen shows the pallet SSCC, the number of boxes and «Напечатано» (Printed). Apply the label to the pallet. The handheld then waits for the first box of the next pallet.



Handheld: the pallet is closed and its label printed

Step 6. If the label did not print

On a print error the handheld says «Этикетка не напечатана» (Label not printed) and the reason. Fix it and press «Повторить» (Retry), or choose «Другой принтер» (Another printer). «Отложить этикетку» (Postpone label) leaves the label for later: the main screen gets a line about unprinted labels that opens a list with «Напечатать все» (Print all). If the handheld says «Результат печати неизвестен» (Print result unknown), look at the printer: if the label came out, press «Этикетка напечаталась» (The label printed), otherwise «Напечатать ещё раз» (Print again).



Handheld: the printer does not respond

Above the closure result, the screen names the printer recorded for this pallet. Earlier versions could show «Не назначен» (Unassigned) even when printing was configured; older pallet data may also lack the name. Check the «Паллета» (Pallet) assignment in settings and the actual label at the printer.

Important: If the host did not accept a box, the «Паллеты» (Pallets) screen shows «Снимите с паллеты N короба» (Remove N boxes from the pallet) with the boxes and the reason. Remove those boxes from the pallet and press «Принято» (Understood); if the pallet is already closed and labelled, press «Перепечатать этикетку» (Reprint label) and replace the label.

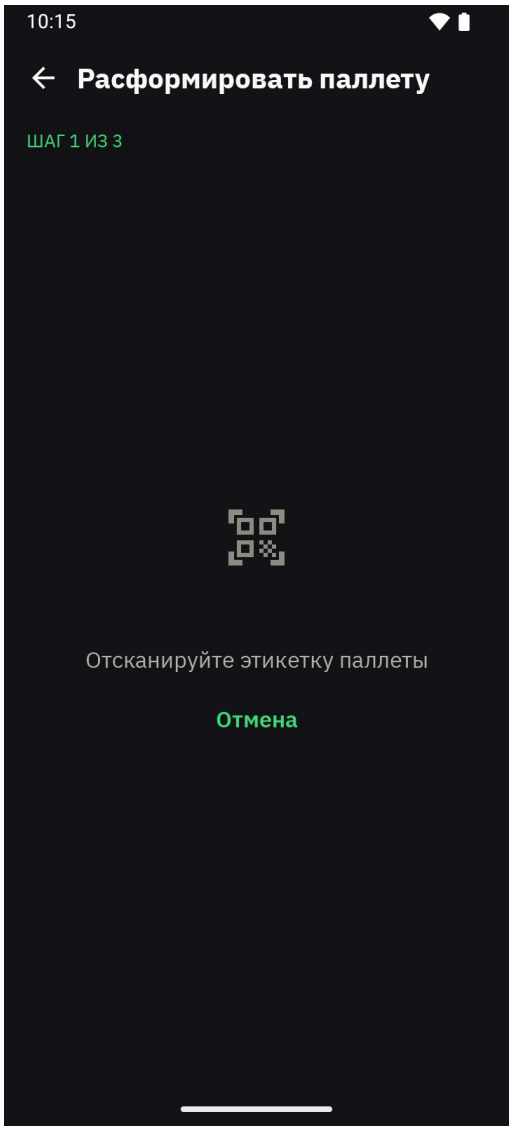
7. Disassembling a warehouse pallet

Important: Follow this procedure only after a coordinated handheld and host update: transmitting disassembly requires host protocol 8 or later. Confirm compatibility with the administrator. Before the update, local disassembly may not reach the host or release the boxes in its records.

This procedure applies only to your own closed warehouse pallet, built on this handheld. Before the operation, confirm with the administrator that its closure has been acknowledged by the host. Production-shift pallets, box exceptions and pallet reprint exceptions are outside this procedure: the handheld does not yet transmit them to the host.

Step 1. Scan the pallet

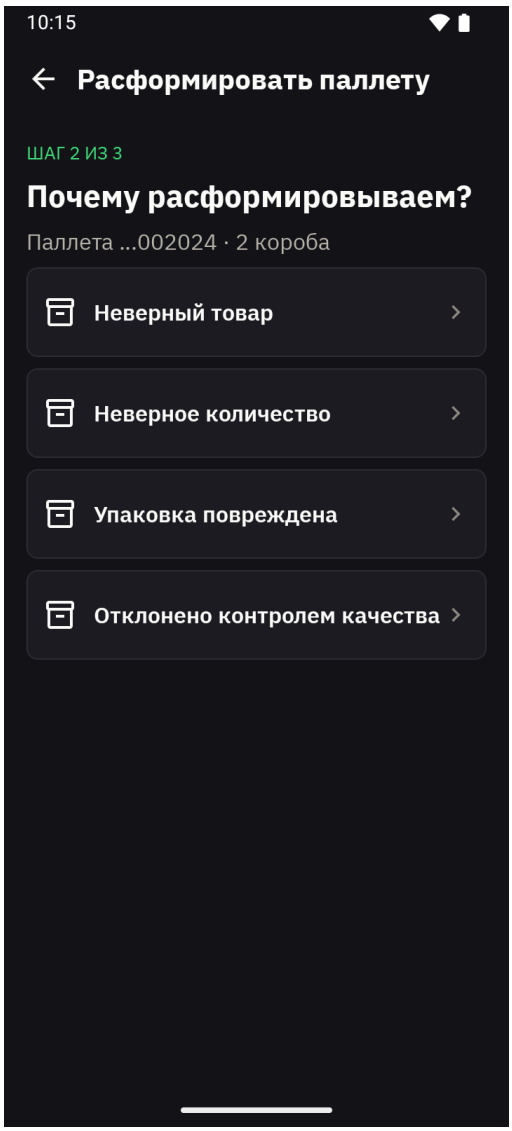
Open «Паллеты» (Pallets) and press the layers icon, «Расформировать паллету» (Disassemble pallet), in the top bar. Scan your own closed warehouse pallet's label. Even if the screen offers a production-shift pallet, do not use this procedure for it.



Handheld: scanning the pallet for disassembly

Step 2. Choose a reason

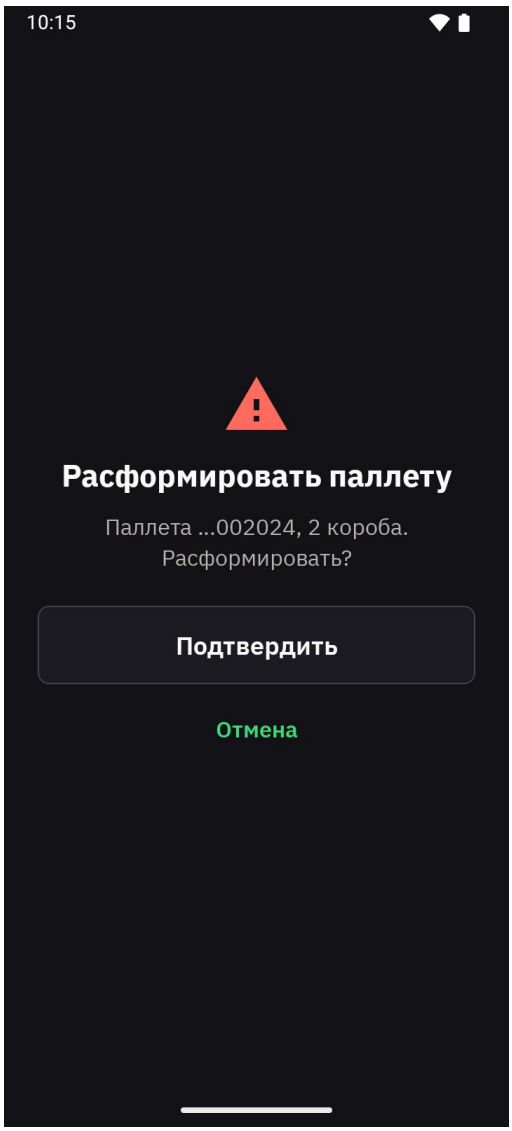
On «Почему расформировываем?» (Why disassemble?), choose the appropriate reason from the list. The pallet's final SSCC digits and box count appear above it.



Handheld: warehouse pallet disassembly reason

Step 3. Confirm and check receipt by the host

Check the SSCC's final digits and box count, then confirm disassembly. After «Паллета расформирована» (Pallet disassembled), remove the old label. This message confirms the local operation: the host releases the boxes after accepting the record. Wait for a host connection and check receipt with the administrator; an empty queue immediately after the operation does not yet prove receipt. If the record is rejected in «Очередь хоста» (Host queue), pass it to the administrator. On other handhelds, then press «Обновить реестр» (Refresh registry) before building a new pallet from these boxes.

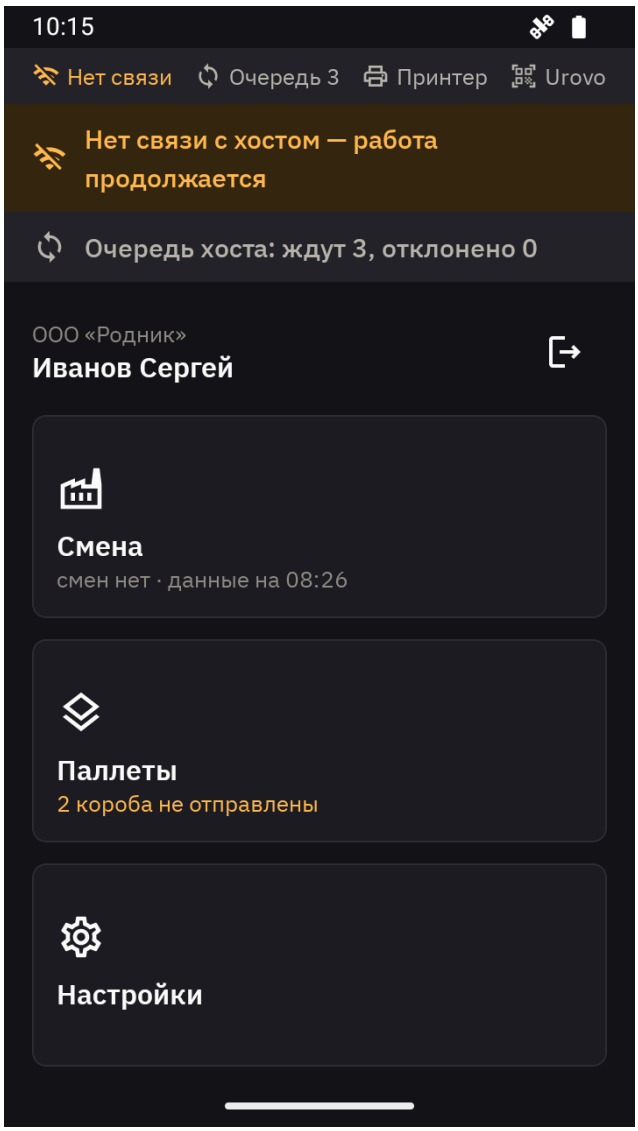


Handheld: confirming the SSCC and box count

8. Working without a link to the host

Step 1. Keep working

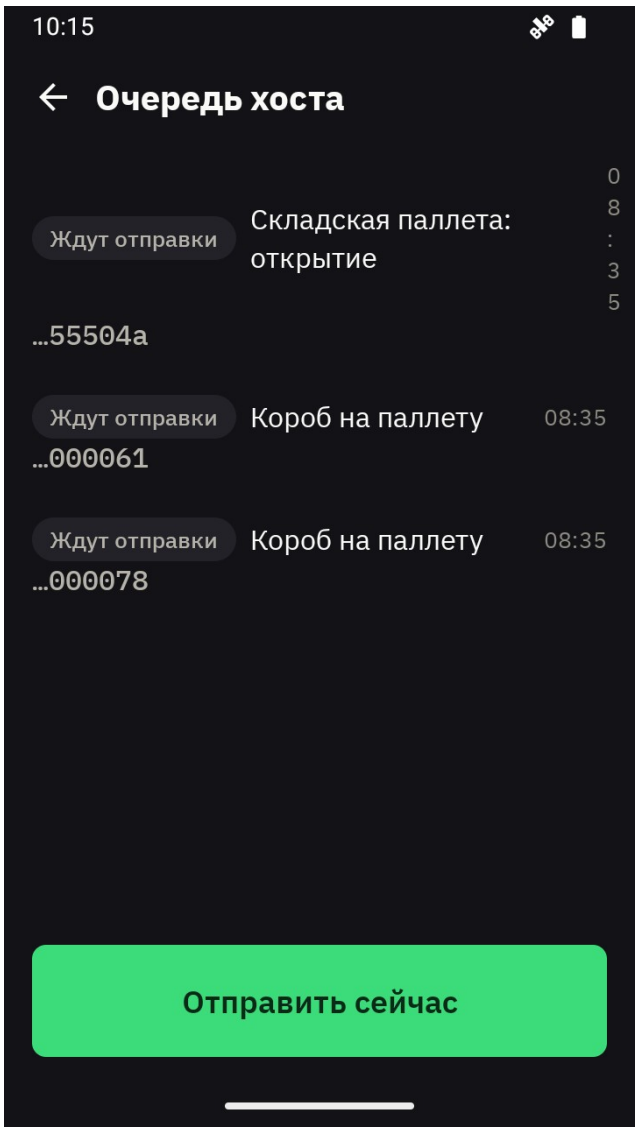
Without a link the handheld shows «Нет связи с хостом — работа продолжается» (No link to the host — work continues). Only boxes from the previously downloaded registry are available; closing pallets requires remaining SSCC numbers. If the numbers run out, reconnect to the host: the pallet remains open. Within these limits, pallet building and label printing go on: records collect on the handheld, and the «Очередь хоста» (Host queue) line shows how many are waiting. The «Паллеты» (Pallets) tile shows how many boxes are unsent.



The handheld without a link to the host

Step 2. Check the queue

Press the «Очередь хоста» (Host queue) line: the «Очередь хоста» (Host queue) screen lists the records that «Ждут отправки» (Wait to be sent) and those «Отклонено хостом» (Rejected by the host). When the link returns, the handheld sends the queue itself; «Отправить сейчас» (Send now) sends it without waiting. If the host rejected a record, press «Передать в поддержку» (Pass to support) and tell the administrator.



Handheld: the queue of records for the host

Important: Do not uninstall the app or reset the handheld while the queue is not empty: unsent records are stored only on this handheld.

9. Common questions

What you see	What to do
The «Смена» (Shift) tile says «смен нет» (no shifts)	Check that the handheld has a line assigned and that the line has planned or active shifts. The administrator chooses «Линия» (Line) in the handheld's window under «Устройства» (Devices) and saves; the handheld receives it when connected to the host (step 3.7). If the field is absent, first update the office PC application. The warehouse «Паллеты» (Pallets) mode also works without an assigned line.
«Справочник ещё не загружен» (Reference data not loaded yet) in «Паллеты» (Pallets)	The handheld has not yet received products and templates from the host. Go back to the main screen and open «Паллеты» (Pallets) again while the handheld is online with the host.
«Сертификат хоста изменился — нажмите, чтобы привязать ТСД заново» (The host certificate changed — tap to pair the handheld again)	The host presents a different certificate. In the office, in «Устройства» (Devices), press «Привязать заново» (Pair again) in this handheld's row; on the handheld tap the message and enter the new code. Data and the queue on the handheld are kept.

What you see	What to do
«Часы устройства расходятся с хостом — проверьте время на ТСД» (The device clock differs from the host — check the time on the handheld)	Set the same time and time zone on the handheld as on the host computer.
«Хост восстанавливается из копии — работа продолжается» (The host is being restored from a backup — work continues)	The administrator is restoring the host from a backup. Keep working: the queue goes to the host after the restore.

Common questions

If a question is not resolved, tell the site administrator: the office shows the handheld, its version and its link to the host in «Устройства» (Devices).